88 % OEE looks fine. It hides 1,108 lost drives.

One electric drive line, 21 shifts. The target is met on paper, yet more than a thousand drives were never built — and nobody saw the next stop coming.

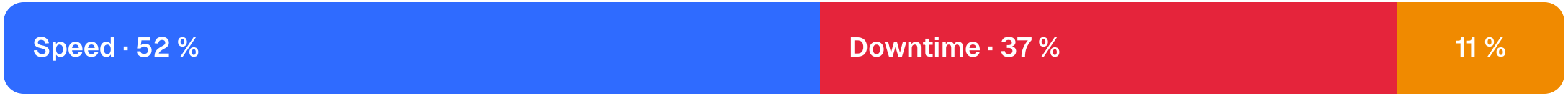
Monthly OEE	88.0% target 85 %
Drives lost	1,108
Warnings before a stop	0

WHERE THE LOSSES ARE

Half of the lost drives are speed, not breakdowns

1,108 drives lost in 21 shifts

1 lost piece = 1 planned minute at the 60 s cycle



Speed

579 pcs

Slowdown 327 · microstops 200 · retests 52 — invisible in the downtime log

Downtime

410 pcs

33 stops; the camera ST-038 alone costs 100 min

Scrap

119 pcs

Fluid leaks Q-031 are the biggest single cause (38 pcs)

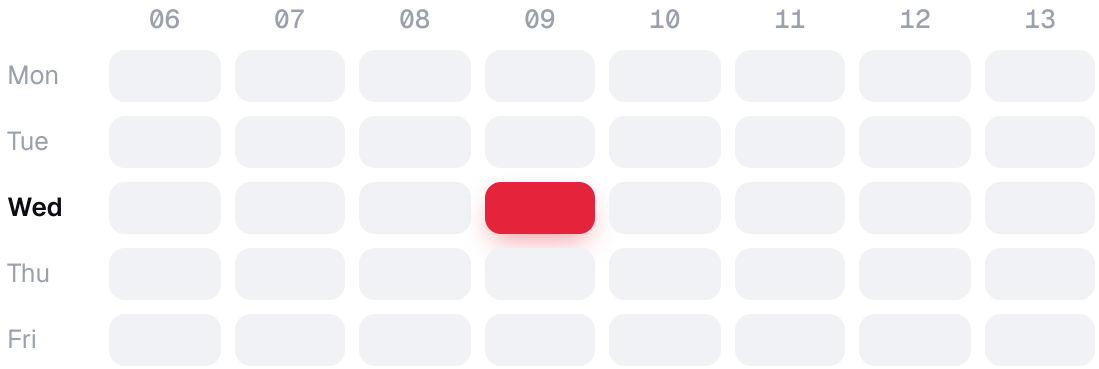
WHAT THE ENGINE FOUND

The losses repeat — so they can be fixed

ST-038 · Camera light module

Every Wed 09:08

4 of 4 Wednesdays, ~25 min each · 100 pcs lost



Why: a scheduled pattern, not random wear — it can be prevented before the shift.

60 — **ST-047 · Test nest leak**
Every Thursday 08:08 · 3 of 3 weeks

105 — **Shift hours 1 and 5 run slow**
88.5 % and 86.9 % speed vs 94.6 % — start-up and after the break

79% — **Three codes make most scrap**
Q-031 leaks, Q-020 press-fit, Q-026 screw torque

Pieces lost in the month (21 shifts); 79 % = share of all scrap

HOW IT WORKS

From line data to the right fix, on the phone

01

Live line data

Hours, stops and scrap streamed every second (MES export today, PLC tomorrow)

02

Detection engine

Finds repeating stops, slow hours and scrap Pareto; predicts the next stop

03

AI assistant

Gemini answers “why” and “what to do” — only from the line data, in English or Slovak

04

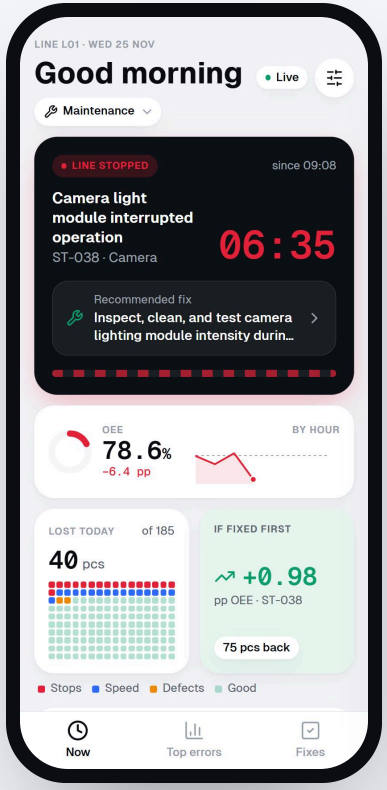
Fix with an owner

82 error codes mapped to Lean countermeasures and a responsible role

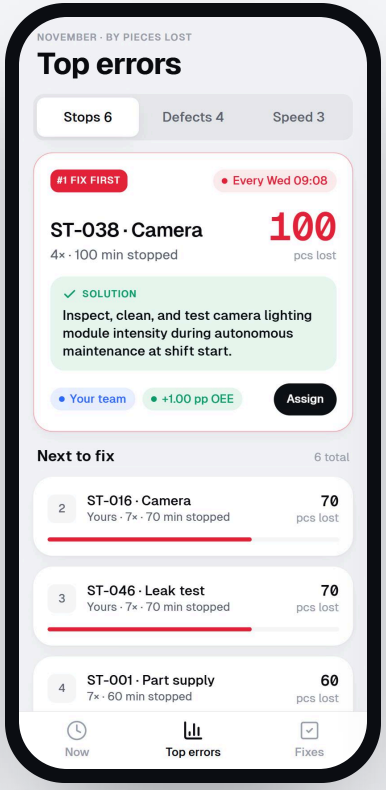
- **Coming up** — Warns **18 min before** a repeating stop
- **If we fix it** — Shows the OEE gain **before** anyone acts
- **Your role** — Maintenance, Quality, Logistics… see **their** fixes first

LIVE DEMO

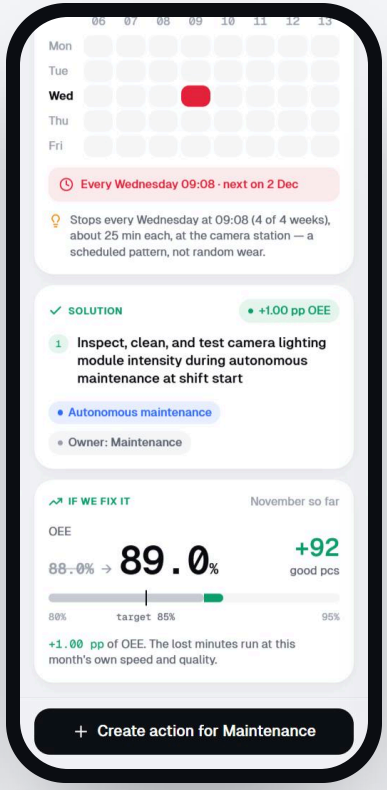
Built for the phone on the shop floor



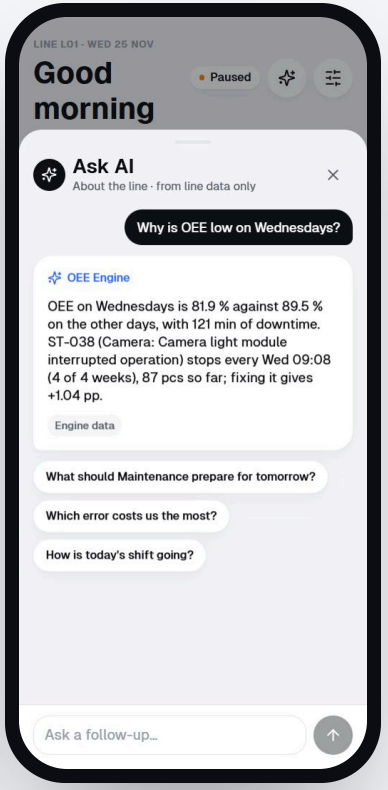
Now · line stopped



Top errors · fix first



If we fix it · +1.00 pp



Ask AI · why?

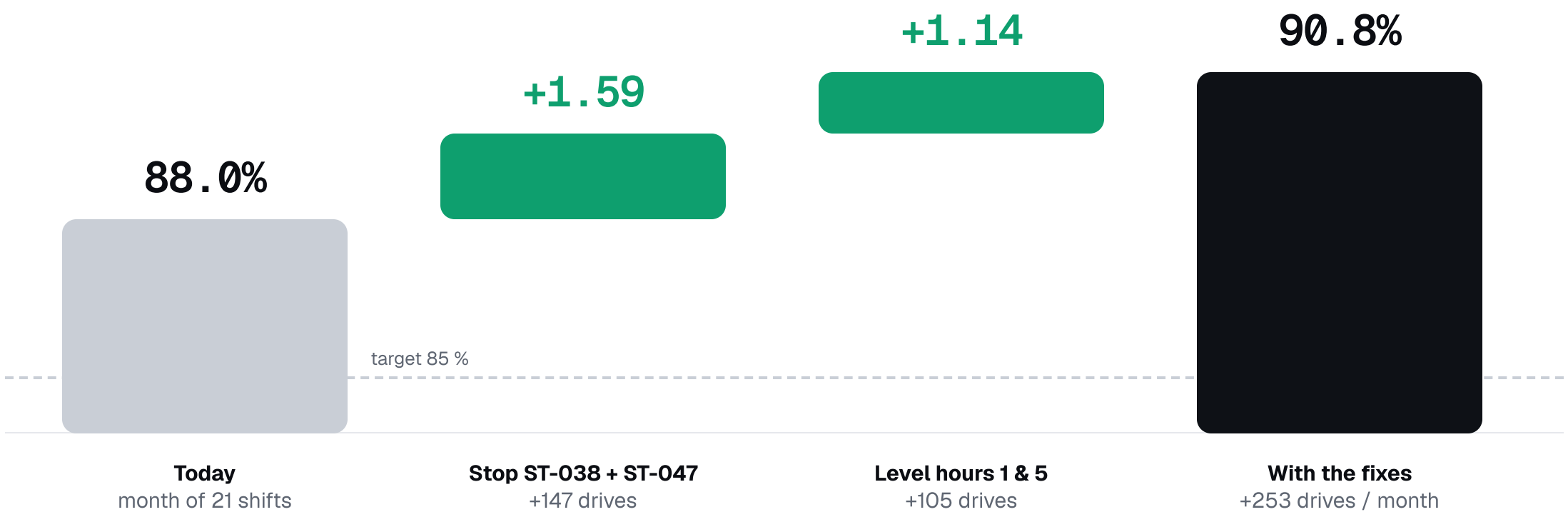
Scan to try it

oee-engine.onrender.com

Installs like an app ·
light and dark theme

IMPACT

+2.74 pp OEE · +253 drives a month · $0 CapEx



What-if model: removed losses run at the month's own speed and quality. Fixes are standard work and autonomous maintenance — no new equipment.

NEXT STEPS

Pilot it on line L01

MONTH 1

Live data on L01

PLC / OPC-UA feed instead of the export; fix ST-038 and ST-047 first

MONTHS 2–3

Push alerts & shift handover

“ST-047 in 30 min” on the right phone; AI writes the handover report

MONTHS 4–6

10 assembly lines

Same engine, plant sign-in, compare lines and share fixes

Every loss visible. Every fix owned. Before the line stops.